Elastic as a Vector Database for RAG-based CTI Tasks

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Welcome

Yury Sergeev | Threat Intel Engineer

I'm a **cybersecurity engineer** specialising in **threat intelligence, security operations**, and **SOC automation**.

Also, I am the **founder of RST Cloud**, an Australian threat intelligence company helping businesses in the US, EU, UK, Middle East, and beyond stay ahead of cyber threats.

Outside of work, I train with the katana, naginata, and other traditional Japanese weapons*.

* Search for **Katori Shinto Ryu** in Sydney

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Agenda

- What is CTI and why retrieval matters
- From human-readable to machine-readable (RST Cloud)
- Threat reports as a knowledge base in Elastic
- Search quality: keyword → vectors → hybrid
- RAG with a local LLM (FastFlowLM example)
- Elastic Agent with internal knowledge base

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What is Cyber Threat Intelligence?

Analysed actionable information regarding known or potential threats

Structured knowledge

About threat actors, malware, campaigns, tools, TTPs, IOCs, vulnerabilities

An enabler

Cyber Management, SOC Analysts, Detection Engineering, Threat Hunting, Incident Response, Risk Management, Vulnerability Management

Sources

Research, vendor reports, blogs, C2 trackers, malware analyses, incident write-ups, honeypots, OSINT, etc

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The problem

Human-readable at scale

Huge volume of PDFs, blog posts, articles, and vendor reports...

Written for humans

Hard to operationalise at scale.

Cannot be manually processed

There are 55 huntable reports out of ~400 TI-related reports a week. Reading everything is almost impossible. Search is often *keyword-only* and misses concepts and synonyms

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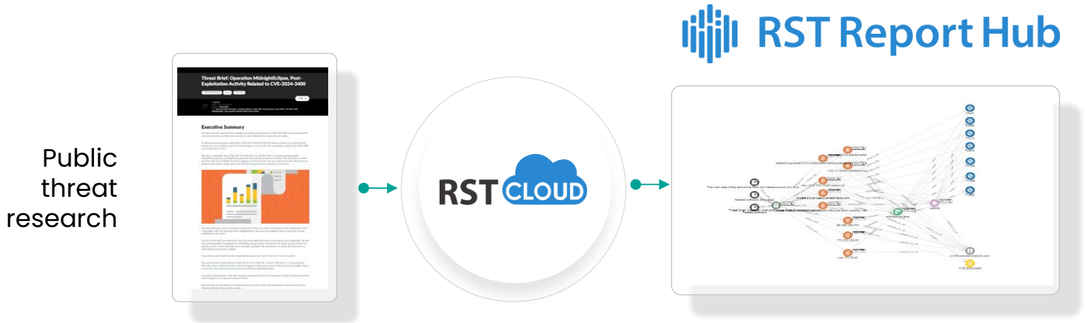
From human-readable to machine-readable

RST Cloud: threat reports → structured data

We turn threat reports into:

- **JSON** metadata: title, summary, IOCs, TTPs, MITRE ATT&CK, categories, dates, content, etc.
- **STIX** bundles: indicators, observables, techniques, threat actors, malware, campaigns, tools, attack-patterns, vulnerabilities, YARA, etc.

Reports become **machine-readable** and **operational**



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What “operationalise” mean

Detection Engineering

Create new behaviour-based rules with insightst from threat intel

Threat Hunting

Query by technique, actor, or campaign

Automation

IoC Feed pipelines, SIEM, and **RAG** with the same corpus

Knowledge base

One place to search all threat reports

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Building a RAG System in Elastic in 15 minutes

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Threat reports as a knowledge base in Elastic

RST Report Hub

Elastic Open Web Crawler

→ **CTI Reports** →

Index with semantic_text fields

Large Language Models are unreliable for Cyber Threat Intelligence. Emanuele Mezzi. **https://arxiv.org/html/2503.23175v1**

Automatic / abstraction

semantic_text

sparse_vector

dense_vector

Index

JSON representation of a report:


```
{
  "content": "## Full report text ..",
  "report_id": "20250117_microsoft_com_report_0x32e6ff7e",
  "report_title": "New Star Blizzard spear-phishing campaign …",
  "report_source": "www.microsoft.com",
  "report_published_date": "2025-01-17",
  "report_url": "https://www.microsoft.com/.../blog/2025/01/16/...",
  "report_categories": ["phishing", "malware", "scan", "c2"],
  "report_threats": ["seaborgium_group", "spear-phishing_technique"],
  "report_mitre_attack_ids": ["T1566.002"],
  "report_industry": ["government", "ngo"],
  "report_geo": ["Russian", "Russia", "Ukraine"],
  "report_idea": “New phishing techniques are...",
  "report_facts": "Russian threat actor Star Blizzard shifted tactics...",
  "report_iocs_domain": ["civilstructgeo.org", "aerofluidthermo.org"],
  "report_observables_domain": ["t.ly"]
}
```


```
{
"feature1":1.2231234,
"feature2":0.3234331,
...
}
```

Most values are zeros, KV that make it interpretable.

Faster, in general works well. No tuning needed

```
[
 0.13586345314979553,
 -0.6291824579238892,
 0.32779985666275024,
 0.36690405011177063,
 ...
]
```

Most values are non-zeros, has a fixed dimension and every number represents some feature that is hard to interpret.

May deliver higher performance, may need a specific model trained

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Demo – Threat Reports Search

1. Import threat reports
2. Search using BM25
3. Sparse vector search
4. Dense vector search
5. Hybrid: BM25 + sparse search

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Demo: Threat Reports + AI

1. Turn the index into a knowledge base
2. Use Elastic AI Assistant to answer based on the knowledgebase
3. Connect with a local LLM
4. Connect with a specialised CTI Agent
5. Build a custom AI agent (Elastic Agent Builder)
   - Detection Engineer Assistant
   - Threat Hunter Assistant
   - SOC Analyst Assistant

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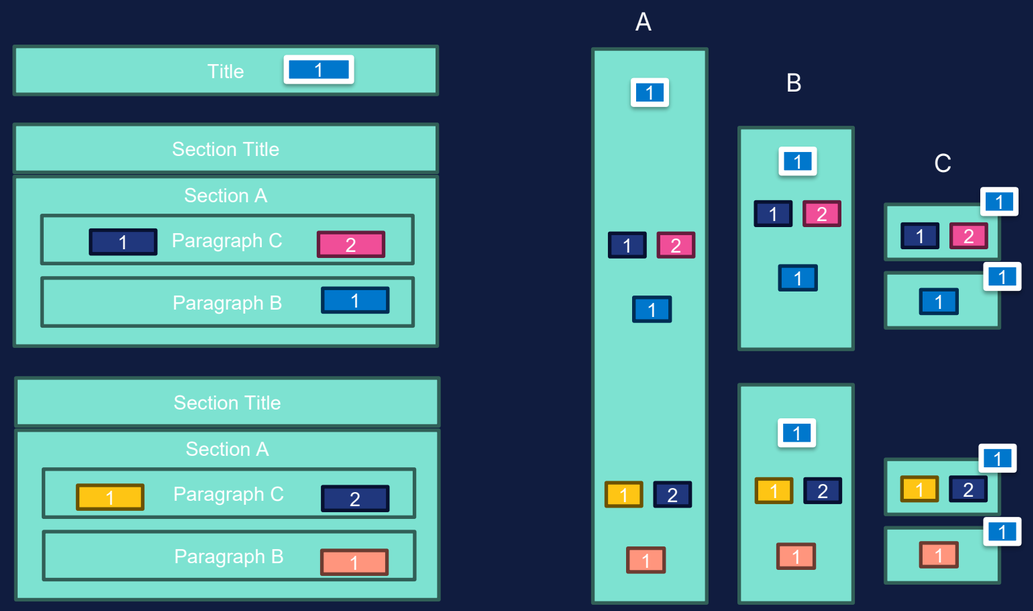
Tokens	Characters
8,782	39745

2024, while the exposure of multiple\nvulnerabilities likely offered the subgroup more access than ever, it appeared\nto have honed its focus to the United States, Canada, Australia, and the\nUnited Kingdom.\n\nThis subgroup s historical pattern of exploitation has also led to the\ncompromise of globally diverse organizations that appear to have limited or no\nutility to Russia s strategic interests. This pattern suggests the subgroup\nlikely uses an opportunistic \"spray and pray\" approach to achieving\ncompromises at scale to increase the likelihood of acquiring access at targets\nof interest with limited tailored effort. In cases where a strategically\nsignificant target is compromised, we have observed significant later post-\ncompromise activity. The geographic focus of the subgroup frequently\ntransitions between broad

Split texts into chunks?

An average threat report is around 9K tokens.

A parsed report with its model is split into chunks with extracted entities



Chunking Strategies

2024 FIRST Cyber Threat Intelligence Conference,
Berlin, Germany, 15-17 April 2024

Processing threat reports at scale using AI and ML: Expectations and Reality
https://www.youtube.com/watch?v=aD6WZWYtn8U

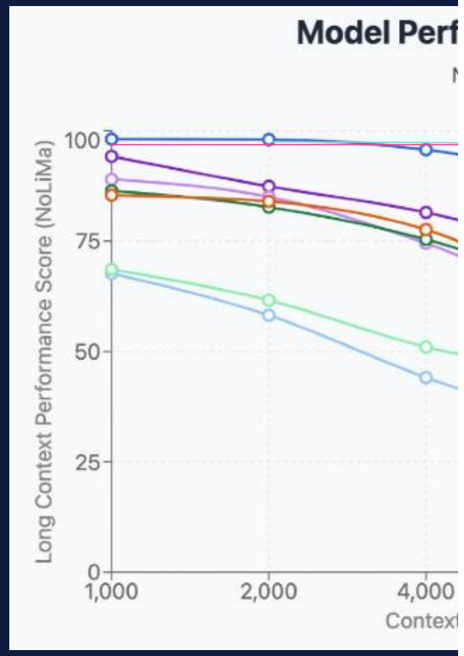
Data Quality

Do we need chunking for Threat Reports?

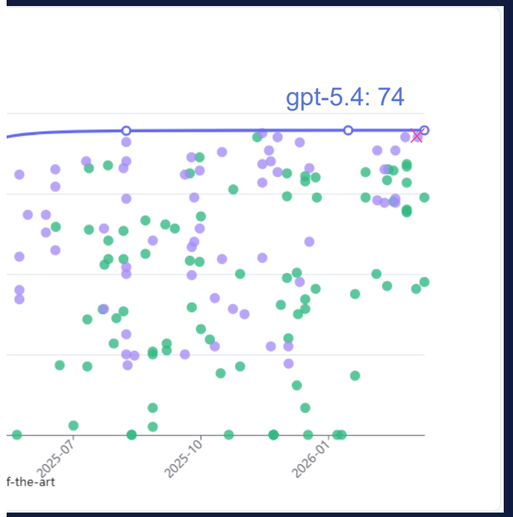
Modern models have long context windows, but the effective window is what matters. Models in 2026 are at a new level.

The original research I referenced when I was speaking on a similar topic at ElasticON 2025

GPT-4o goes from 99.3% to 69.7% wh



#	Model	Params	Context	Date	Overall (%)		Easy (%)		Hard (%)		Short (%)		Medium (%)		Long (%)	
						w/ CoT		w/ CoT		w/ CoT		w/ CoT		w/ CoT		w/ CoT
1	Gemini-2.5-Pro Google	-	1M	2025-03-25	-	**63.3**	-	75.0	-	56.1	-	67.2	-	56.3	-	71.0
2	Gemini-2.5-Flash Google	-	1M	2025-04-17	-	**62.1**	-	72.3	-	55.8	-	68.3	-	60.0	-	55.7
3	Qwen3-235B-A22B-Thinking-2507 Alibaba	235B	256k	2025-07-25	-	**60.6**	-	70.5	-	54.4	-	62.8	-	59.9	-	58.1
4	DeepSeek-R1 DeepSeek	671B	128k	2025-01-20	-	**58.3**	-	66.1	-	53.4	-	62.2	-	54.4	-	59.3
5	Qwen3-235B-A22B-Instruct-2507 Alibaba	235B	256k	2025-07-22	**46.7**	**58.3**	51.6	66.7	43.7	53.1	52.2	63.3	44.2	55.3	42.6	55.6
6	o1-preview OpenAI	-	128k	2024-09-12	-	**57.7**	-	66.8	-	52.1	-	62.6	-	53.5	-	58.1
7	DeepSeek-R1-0528 DeepSeek	671B	128k	2025-05-28	-	**56.7**	-	59.4	-	55.0	-	66.7	-	50.9	-	51.4
8	MiniMax-Text-01 MiniMax	456B	4M	2025-01-15	**52.9**	**56.5**	60.9	66.1	47.9	50.5	58.9	61.7	52.6	56.7	43.5	47.2
9	Gemini-2.0-Flash-Thinking Google	-	1M	2025-01-21	-	**56.0**	-	62.8	-	51.9	-	61.1	-	55.2	-	49.1
10	Human	N/A	N/A	N/A	**53.7**	**53.7**	100	100	25.1	25.1	47.2	47.2	59.1	59.1	53.7	53.7
11	Gemini-Exp-1206 Google	-	2M	2024-12-06	**49.3**	**52.5**	52.9	61.5	47.1	47.1	53.9	55.6	47.1	49.5	45.8	53.3
12	GPT-4o OpenAI	-	128k	2024-11-20	**46.0**	**51.4**	50.8	54.2	43.0	49.7	47.5	59.6	47.9	48.6	39.8	43.5



to 100K
g-context-reasoning

NoLiMa: Long-Context Evaluatio
Munich, Germany. 7 February 2025
https://arxiv.org/abs/2502.05167

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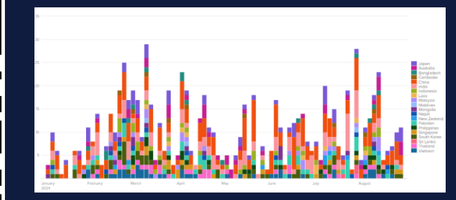
Useful CTI articles on APAC come from

My top 50 sources* from 2025 out of 480+ we monitor

akamai.com
asec.ahnlab.com
bi.zone
blog.f-secure.com
blog.sekoia.io
blog.talosintelligence.com
blogs.blackberry.com
blogs.infoblox.com
blogs.jpcert.or.jp
cadosecurity.com
cisa.gov
cloud.google.com
ctfiot.com
cyber.gov.au
cyberint.com
cybersecurity.att.com
cyble.com
elastic.co
fortinet.com
g0njxa.medium.com
genians.co.kr
group-ib.com
harfanglab.io
hub.dragos.com
jp.security.ntt
labs.k7computing.com
malwarebytes.com
microsoft.com
news.sophos.com
nsfocusglobal.com
orangecyberdefense.com
proofpoint.com
rapid7.com
recordedfuture.com
research.checkpoint.com
resecurity.com
securelist.com
securityintelligence.com
sentinelone.com
seqrite.com
socradar.io
symantec-enterprise-blogs.security.com
sysdig.com
trellix.com
trendmicro.com
unit42.paloaltonetworks.com
uptics.com
welivesecurity.com
wezard4u-tistory.com
zscaler.com

Free Top 10 CTI Reports Weekly Digest
https://medium.com/@rst_cloud



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* Some of the sources are not active in 2026

Summary

- **Different search strategies** should be used depending on the type of user query.
- **Ingesting threat reports** into Elastic helps provide context during investigations, threat hunting, or when building new detection rules
- **AI can provide more reliable and actionable results when it is properly grounded**
- **Automate the basics** – Automation is essential to ensure threat intel workflows run seamlessly and efficiently

Thank you

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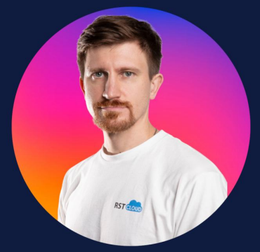

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